Lesson 1 Introduction to ROS2 and Comparison with ROS1

1. ROS2 Overview

ROS2 is the second generation of Robot Operating System, an upgraded version of ROS1, which addressing some of the issues present in ROS1. The earliest version of ROS2, Arden, was introduced in 2017. With iterative updates and optimizations, it has now released the stable versions. Similar to ROS1, the choice between Linux version and ROS2 version is also correlated. The corresponding versions for both are as follows:

ROS2 version	Ubuntu version
Foxy	Ubuntu20.04
Galactic	Ubuntu20.04
Humble	Ubuntu22.04

2. ROS2 Features

1) Distributed architecture: ROS2 adopts a distributed architecture, allowing different nodes to run on multiple computers. This enables ROS2 to operate in larger-scale robotic systems and support higher-level parallel processing and communication.

2) Multi-language Support: ROS5 supports various programming language, encompassing C++, Python, Java, and others. This allows developers to write ROS applications in their preferred language, enhancing development efficiency and flexibility.

3) Enhanced Communication Mechanism: ROS2 introduces a refresh communication system called Distribution Service (DDS). It is a high-performance and real-time messaging protocol that enables implement reliable data communication with ROS2.

4) System-level Tools: ROS2 comes with a set of system-level tools for managing and monitoring ROS2. The tools include management tools (such as Colcon), logging tools (such as Rosout), diagnostic tools (such as Rqt) and others, making it easier for developers to build, test, and debug robotic systems.

5) Real-time Performance: The design of ROS2 takes into account the real-time performance. It provides several real-time performance tools and libraries, such as Real-Time Executor(RTE), Real-Time Publisher(RTPS), and etc. This allows ROS2 to be used in application scenarios where real-time performance requirements are high.

6) Ease of Extensibility and Integration: ROS2 supports a modular architecture, allowing developers to easily add new functions and extend ROS2. Furthermore, ROS2 integrates with other commonly uses roboti software and libraries, such as Gazebo simulator and MoveIt robot motion planning library.

3. Difference Between RO2 and ROS1

3.1 Platform

ROS1 currently only supports running on Linux systems, commonly set up and used on Ubuntu. Meanwhile, ROS2 can currently be set up and used on Ubuntu, Windows and even embedded development boards, making its platform more extensive.

3.2 language

- C++

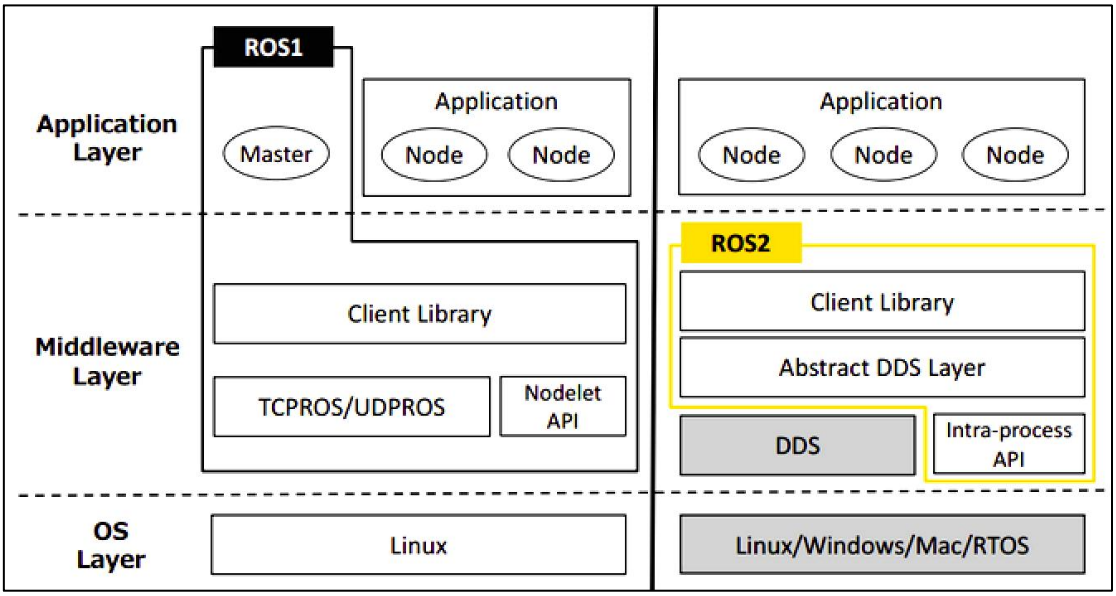
The core of ROS1 is based on C++03, while ROS2 extensively utilizes C++11.

- Python

ROS1 primarily uses Python 2, whereas ROS2 requires Python version 3.5 or higher, with Humble specifically requiring Python version 3.6.

3.3 Middleware

In ROS1, it is necessary to start roscore before initiating, which acts as master node managing communication between all nodes. However, in ROS2, there is no equivalent to roscore; instead, there is only an abstract intermediate interface for data transmission. Currently, all implements of this interface are based on the DDS (Data Distribution Service) standard. This allows ROS2 to provide various high-quality QoS service strategies, improving communication in different network environments.



3.4 Compilation Command

The compilation command for ROS1 is "**catkin_make**," while for ROS2, the compilation command is "**colcon build.**"

If any further development learning is required, please refer to the official tutorial via the following link.

https://docs.ros.org/en/rolling/index.html